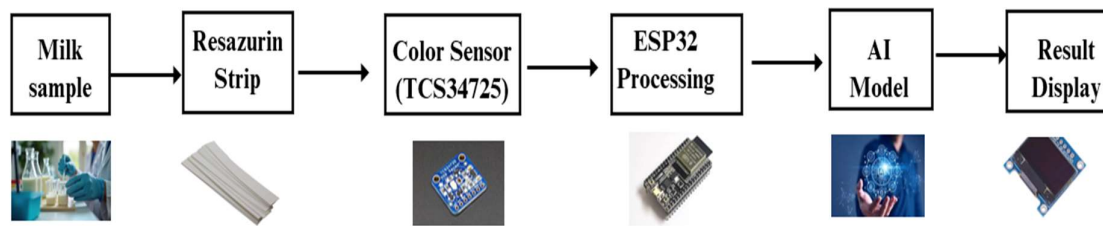


Block-wise Description



Milk Sample

The fresh or stored milk sample is taken and poured into the testing chamber. The milk sample then undergoes a chemical reaction to assess the microbial presence and level of spoilage.

Resazurin Strip

Resazurin is a colorimetric dye used to detect microorganisms in milk.

- Fresh milk → Blue color
- Slightly spoiled milk → Pink color
- Highly spoiled milk → Colorless

The color change is caused by the bacterial metabolism process, which breaks down the color, indicating the level of spoilage.

Color Sensor (TCS34725)

The TCS34725 RGB colour sensor detects colour changes in the resazurin strip.

It provides the intensity values of Red, Green, and Blue colors and then processes them into digital form for further analysis.

ESP32 Processing Unit

The ESP32 microcontroller receives the RGB values from the sensor and undergoes preprocessing tasks such as normalisation and feature extraction.

It also executes the embedded TinyML model for classification.

AI Model (TinyML)

A machine learning model that consumes color variation data is used on the ESP32.

The model is used to classify milk into groups like:

- Fresh
- Moderately Spoiled
- Spoiled

The model works offline without the need for cloud connectivity.

Result Display

The final result of the spoilage is shown by:

- OLED

Colorimetric Response of Milk Samples

